INCH-POUND

MIL-I-3190/7A
8 August 1991
SUPERSEDING
MIL-I-003190/7(SHIPS)
28 July 1969

MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, FLEXIBLE, COATED CLASS 220°C, TYPE E, CATEGORY C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the insulation sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in in that issue of the Department Index of Specification and Standards (DoDISS) specified in the solicitation: MIL-I-3190

The sleeving shall be of the following classification:

Class 220°C, Type E, Category C

REQUIREMENTS:

The insulation sleeving shall be in accordance with tables I, II, and III.

AMSC N/A FSC 5970

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TABLE I. Performance requirements

Property	Conditioning prior to test	Test	Unit	Value required
Dielectric breakdown:				
Straight	96/23/50	4.7.2.2 ASTM D350	Volts (minimum average)	7,000
Straight	96/23/50	4.7.2.2 ASTM D350	Volts (minimum individual)	5,000
Straight	96/23/96	4.7.2.2 ASTM D320	Percent of average value obtained on test after condition 96/23/50 (minimum)	80
Cold brittleness	96/23/50	4.7.4 ASTM D350	Degrees Celsius (maximum)	minus 70±2
Flammability	96/23/50	4.7.5.1 Method A	Seconds (maximum See 4.7.5.1	Self-extinguishing
Fluid Resistance - Xylene - 1, 1, 1 - Tetrafluoroethylene - Jet Fuel - Alkanex (G.E./IMI 9532-1003)	96/23	4.7.6.2 ASTM D374	% increase in maximum average	
Push-back after aging:				
Visual	168/280	4.7.8 ASTM D350	Observation	No cracks or heat rupture in the coating on two or specimens
Dielectric breakdown - straight	168/280	4.7.8 ASTM D350	Volts (minimum average)	6,000
Transformer Oil	48 hrs. immersion @ 105°C	4.7.6.3 ASTM D350	Observation	No disintegration or swelling
Thermal endurance	See 4.7.9	4.7.9 ASTM D350	Temperature index (C°) at 40,000 hrs. minimum	220
Thermal stability	96/280	4.7.10 ASTM D350	Volts (minumum average	6,000
Hydrolytic stability	336 hours over water at 70°C	4.7.7 ASTM D350	Volts: (minumum average) Observation	6,000 volts No disintegration, reversion or tackiness

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TABLE II. Qualification inspection requirements

Examination or test	Requirement	Test	Minimum number of tests for each size and for each condition as specified in table I
Visual examination	3.3, 3.6	4.6	10
Dimensions	3.4	4.7.1	10
Dielectric breakdown: Straight - 96/23/50 and 96/23/96	Table I	4.7.2.2	10
Cold bend test	Table I	4.7.4	As specified in 4.7.4
Flammability	Table I	4.7.5.1 Method A	5
Transformer Oil	Table I	4.7.6.3	3
Push-back after heat aging	Table I	4.7.8	3
Thermal endurance	Table I	4.7.9	See 4.7.9
Hydrolytic stability	Table I	4.7.7	5
Fluid Resistance	Table I	4.7.6.2	3

TABLE III. Quality conformance inspection requirements

Examination or test	Requirement	Test	Conditioning prior to test	Minimum number of tests per sample unit
Visual examination	3.3, 3.6	4.6	96/23/50	As required per sampling
Dimensions	3.4	4.7.1	96/23/50	As required per sampling
Dielectric breakdown: Straight	Table I	4.7.2.2	96/23/50 96/23/96	10 10
Flammability	Table I	4.7.5.1 Method A	96/23/50	5
Push-back after heat aging Method D	Table I	4.7.8	168/280	3
Thermal stability	Table I	4.7.10	96/280	5

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Intended Use: The temperature index for this class of sleeving at 40,000 hours is 220°C. The sleeving covered by this specification is intended for applications in electrical equipment requiring primary insulation having good low temperature flexibility, high moisture and humidity resistance, excellent mechanical protection and abrasion resistance and freedom from silicon.

In applications requiring resistance to the transfer of high molecular structures from the sleeving to insulating oils, a polyester varnish overcoat improves performance. This composite also exhibits ease of cleanup when carbon or graphite particles are prevalent. In addition to meeting the requirements of Table I, it should also be tested to paragraph 4.7.6.4.

In applications requiring good adhesion of silicons, tetrafluoroethylene fluorocarbon polymers or other materials to the sleeving, an investigation should be conducted to determine acceptability.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5970-1065)

Custodians:
Army - ER
Navy - SH
Air Force - 20

Review activities:
Army - AR, AV, EA, MI
Navy - EC
Air Force - 85, 99, 80

User activities:
Army - ME
Navy - MC, OS

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